



Ethical Challenges Associated with the Use of Artificial Intelligence in University Education

Yuri Reina Marín¹ · Omer Cruz Caro¹ · Angelica María Carrasco Rituay² · Katia Araceli Guimac Llanos¹ · Doris Tarrillo Perez¹ · Einstein Sánchez Bardales³ · Judith Nathaly Alva Tuesta³ · River Chávez Santos⁴

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Abstract

The integration of Artificial Intelligence (AI) within higher education has given rise to substantial ethical concerns and challenges, including concerns regarding data privacy, equity in educational resources, and algorithmic biases. These factors have the potential to compromise the integrity of academic processes. In light of this, the study aims to analyze the perceptions of students and faculty members about the ethical challenges associated with the use of AI in universities. The methodology employed was non-experimental quantitative, with the questionnaires being designed according to Luciano Floridi's algorithmic theory. The study's participants 890 university students and 162 faculty members from 21 higher education institutions in Peru. Results indicate that 51.2% of faculty and 47.5% of students expressed concern about data privacy and security. Moreover, a significant proportion of respondents, 61.1% of faculty and 53.5% of students, believed that AI systems lack transparency. These findings highlight the urgent need for regulatory frameworks that promote ethical AI use, ensure equitable access, and safeguard academic autonomy. It is essential for universities to implement concrete measures, including ethical oversight mechanisms, audit protocols, and digital literacy training programs, to maximize the benefits and mitigate the risks of AI in education.

Keywords Ethical challenges · Artificial intelligence in education · Data privacy · Equity in education · Academic integrity

Introduction

AI has revolutionized various aspects of society, impacting areas such as healthcare, transportation, socialization, and education (Knoth et al., 2024; Nagy, 2024). It has emerged as a disruptive tool with great potential in education, offering new opportunities for individualized learning and content creation (Davy et al., 2025; O'Dea et al., 2024). However, research on AI in education reflects a fragile connection with pedagogical perspectives or teaching approaches, particularly in education (Chiu et al., 2023).

Extended author information available on the last page of the article

AI in university teaching provides tools that improve decision-making, modernize institutional systems, and enhance both academic performance and critical thinking (Fadlelmula & Qadhi, 2024; Walter, 2024). It also simplifies academic writing tasks for students (Baig & Yadegaridehkordi, 2024; Sun & Zhou, 2024) and enables personalized learning approaches (Holmes et al., 2022). The adoption of AI supports professors and students by facilitating learning, increasing productivity, and optimizing institutional operations (Kayali et al., 2023).

As part of AI development, ChatGPT has become the leading task guide, online course transcriber, and developer of multilingual educational materials (Román et al., 2023). AI has the potential to improve the learning of university students by providing real-time feedback and revolutionizing teaching methodologies in the long term (Chan, 2023); facilitating study for convenience and receiving feedback without the direct intervention of an instructor (Dakakni & Safa, 2023). Constant innovation enables more informed decision-making, automating processes, and providing predictive analytics on student performance (Ortega et al., 2024). Consequently, AI offers intelligent tutoring tools, automated assessment, and generation of educational materials, reducing the time burden on professors and students through a dynamic approach (Gallent et al., 2023).

However, integrating AI into higher education presents ethical challenges (Lee et al., 2024); such as the authorship integrity and an increase in academic misconduct among students (Duah & McGivern, 2024). These challenges generate ethical dilemmas that universities must address. Many universities have not yet developed institutional policies to regulate AI use and address these challenges (Duah & McGivern, 2024; Gallent et al., 2023). A lack of knowledge and resistance to change has caused many professors to incorporate AI technologies into their academic work (Chiu & Chai, 2020). This is exacerbated by the absence of rules and regulations that protect user privacy and information (Herrera, 2022). This leads to the misuse of technological advances, perpetuating biases and discrimination in society (Caja et al., 2024).

In the context of the present scenario, the paradigm shift in the approach to university education in the era of AI of paramount importance because universities must assume the commitment to take measures to adapt to the new reality and balance it with the protection of essential academic values, such as integrity, equity, and intellectual autonomy (Zawacki-Richter et al., 2019). Furthermore, it is imperative to contemplate the implications of AI's implementation on the privacy of students and the security of educational data (Wang et al., 2024).

Aspects such as the conceptualization of AI literacy have been examined (Černý, 2024), academic dishonesty and plagiarism (Swist et al., 2024), the need for ethical and social approaches in their teaching and the impact of generative AI on university education show challenges of confidentiality and privacy of information (Gallent et al., 2023). Moreover, in critical domains such as academic essays, AI is regarded as a potential threat to the integrity of evaluation processes and the credibility of academic institutions (Sweeney, 2023).

Despite growing interest in the ethical implications of AI in education, empirical research examining these challenges from the direct perspective of stakeholders, particularly students and instructors, remains limited (Bozkurt et al., 2021; Gallent et al., 2023; Holmes et al., 2022; Karran et al., 2024). A preponderance of extant studies have historically prioritized theoretical frameworks or normative considerations, often to the exclusion of empirical evidence drawn from real-world educational contexts (Cotton et al., 2024; Swist et al.,

2024; Tsamados et al., 2022). While countries in Europe, North America, and Asia have begun to develop robust models of AI governance for education, there is a notable lack of data specific to the Latin American region. In this region, cultural, infrastructural, and policy-related variables may affect the ethical deployment of AI technologies in universities (Amén et al., 2024; Ferhataj et al., 2025). Consequently, there is a need for more context-sensitive research to understand how ethical issues such as data privacy, algorithmic bias, and academic integrity are perceived and addressed within these emerging educational environments.

However, the majority of these investigations are literature reviews or theoretical explorations, which reveal a significant gap in empirical research on the ethical challenges arising from the practical implementation of AI in university contexts. Consequently, there are ongoing initiatives to evaluate the comprehension of ethical dilemmas within distinct domains, such as medicine (Al-Ani et al., 2024), and to collaboratively develop ethical frameworks in the realm of educational AI, with a particular focus on privacy, surveillance, and data utilization in education (Swist et al., 2024). Additionally, the erosion of human connection, data privacy concerns, algorithmic bias, and a lack of transparency have been shown to have a substantial impact on academic performance among university students, the density of errors, and the risk of encountering misleading information (Ortega et al., 2024).

In the context of Peru, the integration of AI in higher education is still at an early stage of development (Reina et al., 2025). While certain institutions of higher education have initiated the integration of AI tools such as ChatGPT and automated learning systems, the implementation of these technologies is frequently unregulated and lacks comprehensive ethical guidelines (Román et al., 2023). The majority of institutions have not yet adopted explicit policies to address fundamental issues such as data protection, transparency of algorithms, or equitable access to AI-based resources (Alonso-Rodríguez, 2024; Herrera, 2022). This issue is further exacerbated by the presence of disparities in digital infrastructure, the limited familiarity with AI among faculty and students, and the dearth of national frameworks that provide guidance on the ethical use of AI within academic settings (Barrios, 2023). Consequently, there is an increasing concern among education stakeholders regarding the risks associated with the unregulated adoption of AI in universities. These risks include the erosion of academic autonomy, threats to privacy, and the perpetuation of inequality (Caja et al., 2024).

There is a necessity for research that directly examines the experiences and perceptions of stakeholders involved in the use of AI in higher education (Dakakni & Safa, 2023). This study aims to address this lacuna by offering empirical evidence regarding the ethical challenges encountered by students and faculty when utilizing AI tools within the university setting. The objective of the present study is to furnish a comprehensive overview of the perceptions held by students and faculty members with regard to the integration of generative AI in university education. This includes understanding their knowledge and utilization of AI tools, their perceived benefits and concerns, and their views on the urgency of establishing AI governance frameworks in universities. To this end, a survey was conducted among university students and faculty from various institutions across Peru. The objective of the survey was to ascertain their ethical perceptions regarding the utilization of AI in academic settings.

Theoretical Framework

The integration of AI in educational settings is escalating rapidly, with the goal of improving education quality and refining teaching methods (Gallent et al., 2023; Sweeney, 2023). As a component of the development of generative AI (Gen AI), it is becoming increasingly present in the field of education, particularly in personalized teaching (Dakakni & Safa, 2023). Its deployment in higher education typically encompasses three core domains: teaching and learning, administrative management, and student assessment (Al-Zahrani, 2024). However, there are several challenges that require immediate attention including school readiness, instructors AI skills, student literacy, and ethics (Kit et al., 2025). Additionally, AI exhibits a superior capacity to deliver standardized content and assessments in an uninterrupted manner, devoid of fatigue (Kayali et al., 2023).

Despite these benefits, the application of AI in higher education raises significant ethical concerns that demand critical scrutiny (Cotton et al., 2024; Dabis & Csáki, 2024). These include data privacy issues, as AI systems often collect vast amounts of personal information, potentially leading to misuse and breaches of student confidentiality (Mauti & Ayieko, 2025). Safeguards against unauthorized access are often insufficient (Gupta & Nyamapfene, 2025). Safeguards against unauthorized access are often insufficient (Ferhataj et al., 2025; Mauti & Ayieko, 2025), while the opaque nature of AI algorithms complicates accountability in decision-making processes (Protopapa & Idris, 2025). The lack of transparency and accountability in algorithmic processes further complicates ethical oversight (Protopapa & Idris, 2025).

In assessments, generative AI can compromise academic integrity by facilitating plagiarism and diminishing opportunities for the development of critical thinking (Kamali et al., 2024; Protopapa & Idris, 2025). AI tools can disrupt traditional tutoring roles, especially in culturally rich educational contexts (Abdelwahab, 2024; Mauti & Ayieko, 2025). Nonetheless, some scholars argue that the advantages, such as enhanced learning outcomes and improved administrative efficiency, may outweigh the risks, provided robust ethical frameworks are established (Muwagunzi et al., 2024).

A recurring theme in the literature is the human dimension of teaching, particularly empathy, which distinguishes educators from AI systems (Amén et al., 2024). Despite this difference, the use of smart tutors and virtual assistants is becoming more frequent because they offer immediate feedback and continuous support through the creation of content such as images, videos, audio, and conversations (Khatri & Karki, 2023). Even so, AI-based platforms such as ChatGPT and Magic School have gained popularity among instructors for their utility in lesson planning and instructional material development (Summers et al., 2024). Other tools like DALL-E and Gemini assist in the creation of visual content and literature reviews, respectively (Barus et al., 2025). Despite these developments, the use of AI remains largely supplemental, as its integration into pedagogical practice is still limited (Guan et al., 2025). This partial adoption has sometimes led to the inappropriate generation of academic content (Paguay et al., 2024).

ChatGPT, in particular, has been recognized as an educational facilitator due to its capacity to support autonomous learning and its integration into digitally enriched learning environments (Saleem et al., 2024). Technologies such as AI-powered automation and content adaptation allow for the personalization of academic experiences, improving both accessibility and learning efficiency (Miao et al., 2021; Aydınlar et al., 2024). These platforms

offer students free access to information and assistance with academic tasks, contributing to a more responsive learning environment (Ortega et al., 2024). In this regard, Singh's "Teach AI" global initiative promotes a socio-ecological model that addresses the training needs of both educators and students (Kit et al., 2025).

Nevertheless, AI algorithms are not immune to embedded biases, as they reflect the perspectives and assumptions of their developers (Kit et al., 2025). This underscores the need for faculty members to rethink their instructional roles and pedagogical strategies to mitigate tensions and foster constructive engagement with AI tools (Saleem et al., 2024). Eliminating resistance to technological change remains a key prerequisite for effective AI adoption (Guan et al., 2025).

Although the benefits of AI in education are evident, their realization depends on a strong ethical foundation that ensures responsible use and respects human well-being (Ortega et al., 2024). AI has proven useful in domains such as language learning and scientific writing, offering assistance with grammar, argument development, and sentence construction through its creative capabilities (Kit et al., 2025). In the health sciences, AI tools have demonstrated utility in both patient education and simulated learning for healthcare professionals (Clemente et al., 2024; Arjanto & Aditama, 2025). Despite its potential, AI cannot replace the originality and critical insight that characterize scientific research (Barrios, 2023), and it often produces fictitious citations, which limits its reliability (Annamalai et al., 2025).

The ethical use of AI in higher education must also address concerns related to academic integrity, such as plagiarism and authorship (Černý, 2024; Guan et al., 2025). AI-generated responses and summaries, while convenient, may be misunderstood or misapplied by students who lack a deep understanding of the subject matter (Annamalai et al., 2025; Bozkurt et al., 2021). Therefore, it is necessary to establish clear and regulatory academic standards on the use of AI in the search for information, production of research, and academic content, ensuring that students develop writing skills and anthropomorphization is avoided (Gómez et al., 2024; Radtke & Rummel, 2025). Simultaneously, the protection of personal data on AI-powered platforms is a critical concern (Al-Zahrani, 2024). According to Mora et al. (2023) the use of AI for personalized learning involves the mass collection of data, which raises the risk of unauthorized access and unintentional disclosure. To mitigate these risks, universities must implement secure data management protocols (Chen et al., 2020; Ramos & Salazar, 2025).

Luciano Floridi's theory of algorithmic ethics offers a robust theoretical foundation for addressing these concerns. The model emphasizes ethical audits as a means to certify AI reliability and to strengthen user control over automated decision-making systems (Mökan-der & Floridi, 2021). According to Tsamados et al. (2022), Key principles include responsibility, fairness, and prudence. On the other hand, Martin (2019), refers to a sensorial process which is defined as a continuous and productive process that is aligned with ethics from a systematic representation and with the most recent regulatory policies. On the other hand, Henry and Oliver (2022) conceptualize ethics as a structural mechanism capable of confronting the challenges posed by AI, and Aydınlar et al. (2024) assert that ethical engagement is essential for effective AI governance.

However, a recurrent challenge pertains to the necessity for clarity and commitment, particularly concerning algorithms that are inherently grounded in values (Bozkurt et al., 2021; Kraemer et al., 2011). These algorithms are not impartial, but rather engender moral consequences, thereby reinforcing or undermining ethical principles and, in certain instances,

impeding the rights and dignity of the affected parties (Martin, 2019). Consequently, educational systems must be structured in such a way as to prevent the emergence of situations that may harm certain groups (Abisheva et al., 2024). A compelling practice model for the integration of AI in the education sector is the application of pedagogical heuristics, such as the effective use of chatbots based on AI (Sweeney, 2023). For instance, ChatGPT, a leading example of such technology, has the potential to transform communication in classrooms through flexible solutions, thereby promoting continuous improvement and personalized learning experiences. This, in turn, can foster inclusion and equitable access to knowledge (Saleem et al., 2024).

The pressing necessity for research that addresses the experiences and perceptions of AI stakeholders in higher education, including parents, professors, and students, remains a priority (Dakakni & Safa, 2023; Karran et al., 2024). However, there is a paucity of empirical research that addresses the aforementioned ethical concerns, including data privacy, algorithmic bias, and transparency in AI systems utilized within academic institutions (Alonso-Rodríguez, 2024). In this regard, Radtke and Rummel (2025) underscore the necessity for the development of more comprehensive studies on practices and policies that ensure the ethical use and implementation of AI. As AI becomes increasingly integrated into our daily lives, education emerges as a critical factor influencing social well-being (Engelmann et al., 2024; Karuppal, 2025).

The present study makes a significant contribution by addressing the existing gap in research on the ethical implications of generative AI in higher education. Through a focus on the perceptions of university students and professors across Peru, the research identifies knowledge, uses, and concerns regarding these technologies. It also establishes a basis for the development of ethical governance frameworks within each university. In a context where the integration of AI in higher education is advancing rapidly, it is imperative to understand the experiences of the actors directly involved. This understanding is essential to ensure that these tools enhance learning equitably and responsibly, preserving essential educational values and adequately preparing students for a future where AI will be a component of university development.

While the algorithmic ethics approach proposed by Floridi has been influential in shaping the debate on the moral responsibility of intelligent systems, several studies have proposed relevant critiques and extensions that strengthen its application in educational contexts. Mittelstadt (2019) posits that the principles formulated by Floridi are necessary but insufficient, as they tend to privilege transparency without adequately considering the institutional dynamics and power conflicts that affect the development and implementation of algorithms, particularly in sectors such as higher education. Hagendorff (2020) reinforces this criticism by pointing out that many ethical frameworks are excessively abstract and lack effective mechanisms for their practical application, particularly in environments with limited regulatory capacities or hierarchical institutional cultures. Conversely, Morley et al. (2020) propose an evolution of Floridi's framework towards participatory ethics, emphasizing the importance of integrating principles of epistemic justice and deliberative processes that include the actors affected by algorithmic decisions.

Nevertheless, Floridi's model is a suitable structural foundation for this type of study. Firstly, it provides a clear existing framework on the moral agency of AI systems, which allows for the analysis of how responsibilities are distributed in complex institutional contexts such as universities (Floridi & Cows, 2019). Secondly, its proposal is articulated with

principles that have been widely accepted by multilateral organizations such as the OECD and UNESCO, thereby facilitating its integration into research that seeks to propose public policy recommendations. In conclusion, the adaptability of Floridi's approach enables its integration with more contextualized critical frameworks, such as those discussed in this context, without compromising analytical coherence. As Gunkel (2012) and Fjeld (2020), have demonstrated, the strength of the Floridi framework does not lie in its normative rigidity; rather, it lies in its capacity to engage in dialogue with other approaches and to serve as a basis for ethical evaluation in scenarios where specific regulations do not yet exist.

Methodology

The present study adopted a non-experimental quantitative design, as its objective was to analyze and describe the variables in their natural context, without manipulating them. This approach aligns with the methodology outlined by Hernández-Sampieri & Mendoza (2019).

Information gathering

In this study, the research team developed and validated two questionnaires, employing rigorous methodological criteria in their design.

- Questionnaire design based on Luciano Floridi's Algorithm Ethics Theory, which addresses the main ethical challenges in the development and application of AI systems. It was composed of six dimensions (Privacy and data protection; Transparency and explainability; Accountability and academic freedoms; Equity and non-discrimination; Well-being and social benefit; and training and literacy in AI). Each dimension was operationalized into 3 to 4 Likert-scale items, resulting in a total of 23 items per questionnaire. A 5-point Likert scale was used, where responses were coded as follows: (1) Never, (2) Almost never, (3) Sometimes, (4) Almost always, and (5) Always. This structural decision is consistent with established psychometric principles in scale construction, particularly in exploratory research contexts where content coverage and reliability are balanced against instrument length. According to DeVellis (2017) and Boateng et al. (2018), employing a minimum of three to five items per construct ensures sufficient representation of the latent dimension, facilitates internal consistency measurement, and allows for meaningful item-level variance. In the present study, this approach enabled the development of a compact yet theoretically robust instrument aligned with Floridi's algorithmic ethics framework, optimizing the balance between empirical validity and participant response burden.
- To ensure content validity, the instrument was submitted for expert review by three university-level professionals with advanced degrees in education and experience in digital ethics. These experts assessed the items using a structured validation sheet based on criteria such as clarity, internal coherence, objectivity, and relevance to the construct. Each item was reviewed against indicators of construct alignment, risk of bias, and appropriate language for the target population.

- Validation using Cronbach's Alpha: A score of 0.926 was obtained for the student questionnaire and 0.911 for the professor questionnaire. These scores indicate a high level of internal consistency and reliability of the instruments.

The validated instruments were implemented on the Google Forms platform and structured into three sections. The first section was for written informed consent. The second section was for general information. The third section contained questions related to ethical challenges across the six evaluated dimensions. Data was collected by disseminating a survey link via institutional email and through the official document submission offices of all Peruvian universities. To facilitate dissemination, the invitation was sent to university authorities, who forwarded the link to students and faculty through their respective institutional channels.

Participants

A total of 21 Peruvian universities participated in this study, with 14 being public institutions and 7 private. Of these, 890 students and 162 professors from various universities elected to participate.

The sample was chosen by convenience, only professors and students who were willing to partake in the study and had provided their informed consent to divulge information about the study were considered. Participants provided written informed consent to partake in this study. Informed written consent was obtained from the subjects prior to the publication of any potentially identifiable data in this article.

Data Analysis

Data analysis was performed using SPSS software version 26. Descriptive statistics were used to summarize the baseline characteristics and responses of the participants. Categorical data were presented as frequencies and percentages.

To facilitate a clearer interpretation of participants' responses across the six dimensions, the original 5-point Likert scale (1 = Never, 5 = Always) was recoded into a three-level ordinal scale: low, moderate, and high. This transformation was carried out using a conversion formula designed to distribute the possible scores evenly across the three levels, based on the number of items per dimension (n) and the range of possible scores.

The procedure was as follows:

Items with lower score (1).

Items with the highest score (5).

Number of items (n) (varies by dimension).

Rank (k)

$$k = \frac{(n) * (\text{items with highest score}) - (n) * (\text{items with lower score})}{\# \text{ of levels } (3)}$$

For the levels we have:

Low level:

Lower rank = $(n) * (\text{items with lower score})$.

Upper rank = $(n) * (\text{items with lower score}) + k$.

Moderate level:

Lower rank = $(n) * (\text{items with lower score}) + k + 1$.

Upper rank = $(n) * (\text{items with lower score}) + 2k + 1$.

High level:

Lower rank = $(n) * (\text{items with lower score}) + 2k + 2$.

Upper rank = $(n) * (\text{items with higher score})$.

This methodological decision was made to group the continuous scores into broader categories, thereby improving the synthesis and visualization of results across diverse dimensions. The three-level scale facilitates the identification of trends and patterns, particularly in the context of communicating findings to non-technical audiences or stakeholders. It is important to note that this transformation does not eliminate the richness of the data; rather, it enhances interpretability in the context of aggregated analysis and comparative assessment between groups.

Results

To understand the profile of the actors involved in this research, the sociodemographic characteristics of the university students and professors who participated in the study are presented below. This information allows for an appropriate contextualization of the perceptions collected regarding the ethical challenges of using AI in higher education. Tables 1 and 2 describe the variables associated with students and professors, respectively, considering aspects such as gender, generation, type of university, academic field, and, in the case of students, their current academic term.

The surveys were administered to students from the first to the twelfth semester, with 16% of respondents enrolled in the fourth semester, 13.7% in the tenth semester, 11% in the second semester, 10.7% in the eighth semester, and 10.6% in the first semester. The remaining semesters exhibited a participation rate of less than 10%. The data indicates that 52.2% of the respondents identified as female, while 47.8% identified as male.

An overwhelming majority of 92.8% of the students surveyed are part of Generation Z, a generation defined as individuals born between 1997 and 2012. This generation has been raised in a technologically-driven environment, which has engendered a high level of familiarity with digital instruments and AI-based applications such as ChatGPT. The generational classification employed in this study adheres to the criteria established by Daniel Colombo.

The professional careers most represented by the surveyed students are as follows: The field of engineering constitutes 30.9% of the total, followed by the social sciences, commercial education, and law at 26.9%. Health and social services represent 26.7%, while education and services specialties account for 7.3%. Finally, the humanities constitute less than 1% of the total. The classification of these specialties was conducted in accordance with the classification of higher education and productive technical careers established by the National Institute of Statistics and Informatics (INEI), 2024.

With respect to the faculty, 64.2% of professors are male, while 35.8% are female. Furthermore, 40.1% of the surveyed professors belong to Generation X, as classified by Daniel Colombo's taxonomy of human generations (born between 1965 and 1980). This generation

Table 1 Sociodemographic profile of participating university students

Variables	Frequency (890)	Percentage
Gender		
Female	465	52.2
Male	425	47.8
Generation		
Generation Z (1997–2012)	826	92.8
Generation Y or Millennials (1981–1996)	58	6.5
Generation X (1965–1980)	6	0.7
Type of University		
Public	14	33.3
Private	7	66.7
Academic Field		
Education	65	7.3
Engineering	275	30.9
Social sciences, business, and law	239	26.9
Sciences	9	1.0
Health and social services	238	26.7
Services	63	7.1
Humanities and arts	1	0.1
Academic Term		
I	94	10.6
II	98	11.0
III	58	6.5
IV	142	16.0
V	53	6.0
VI	74	8.3
VII	85	9.6
VIII	95	10.7
IX	38	4.3
X	122	13.7
XI	20	2.2
XII	11	1.2

witnessed the emergence of computers and the Internet. The demographic composition of the sample is as follows: 30.2% of the respondents belong to the Y generation, also known as Millennials, who came of age in the digital age and have firsthand experience with the transition to modern technology. 24.1% of the respondents belong to the Baby Boomer generation, who have witnessed the growth of traditional technologies and the initial advancements in digital communication.

The specialties most represented by the surveyed professors are as follows: Engineering represented 39.5% of the total, while social sciences, business education, and law accounted for 29.6%, and health and social services comprised 13.6%. Education and services 7.4% and 3.7% respectively. finally, the specialties of sciences, humanities and arts and agriculture/veterinary medicine with a representation of less than 5% of the total. The classification of these specialties was conducted in accordance with the Higher Education and Technical-

Table 2 Sociodemographic profile of participating university professors

Variables	Frequency (162)	Percentage
Gender		
Female	104	64.2
Male	58	35.8
Generation		
Generation Z (1997–2012)	9	5.6
Generation Y o Millennials (1981–1996)	49	30.2
Generation X (1965–1980)	65	40.1
Baby Boomers (1946–1964)	39	24.1
Type of University		
Public	14	33.3
Private	7	66.7
Academic Field		
Education	12	7.4
Engineering	64	39.5
Social sciences, business, and law	48	29.6
Sciences	5	3.1
Health and social services	22	13.6
Services	6	3.7
Humanities and arts	5	3.1
Agriculture/Veterinary	1	0.6

Productive Career Classifier of the National Institute of Statistics and Informatics (INEI), 2024.

To understand perceptions on how to address the main ethical challenges in the development and implementation of AI systems in universities, a survey was conducted among university faculty and students from all over Peru. These dimensions facilitate a thorough examination of the ethical challenges associated with the implementation of AI systems in universities, ensuring that their development and application are responsible and aligned with the ethical principles proposed by Floridi.

Privacy and Data Protection

This dimension explores the concerns and regulations associated with the management of personal data within the context of the university environment. The study analyzes the manner in which AI systems collect, store, and utilize information from professors and students. It also explores strategies to ensure the security and confidentiality of this data. This section also includes aspects related to informed consent and the rights of access, rectification, and deletion of data. Figure 1 describes these perceptions of university stakeholders.

This dimension focused on the concerns surrounding the collection, storage, and utilization of personal data from professors and students by AI systems. As shown in Fig. 1, 51.2% of professors and 47.5% of students respectively, expressed concerns regarding data privacy and security in the use of AI systems in universities. Of these, 31.6% of students and 18.5% of professors expressed high concern, while 30.2% of professors and 20.9% of stu-

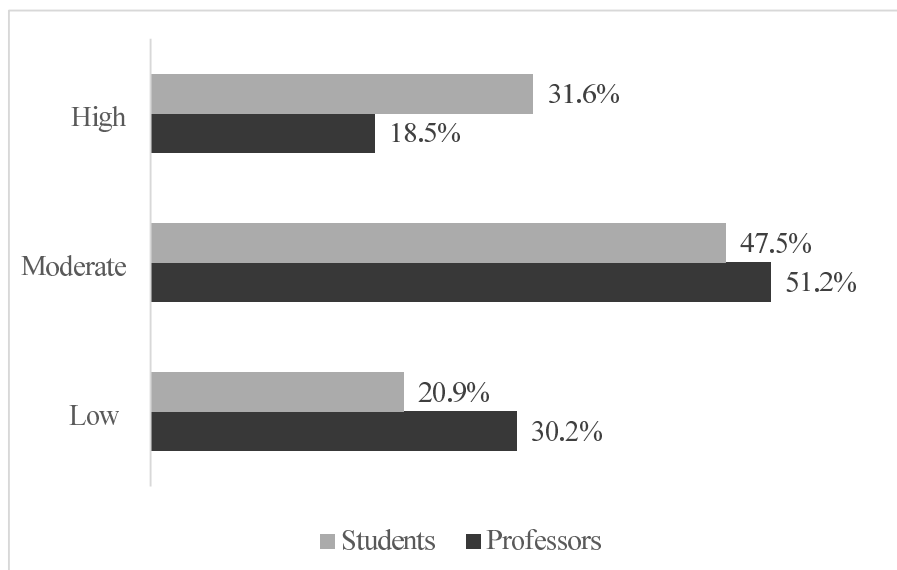


Fig. 1 Perceptions of privacy and data protection. Source: Data obtained from the questionnaire applied

dents indicated low concern. Both groups emphasized the importance of clear institutional policies regulating the collection and processing of personal data. A significant proportion of respondents perceived that universities currently lack effective mechanisms to ensure information protection, underscoring the necessity for regulatory frameworks to mitigate the risks of data misuse.

Transparency and Explainability

This dimension focuses on the need for AI systems to be comprehensible and subject to audit. Transparency involves the clear disclosure of the criteria and processes that algorithms follow for decision-making, while explainability refers to the ability to justify and understand the functioning of these systems. This dimension is key to avoiding hidden biases and promoting trust in AI within the academic field. Figure 2 illustrates the perceptions of professors and students regarding transparency and explainability in the use of AI in universities.

The results show that 61.1% of professors and 53.5% of students moderately perceive transparency, and 36.9% and 17.3% of respondents consider that the AI systems used in the university environment lack transparency in their operation. The absence of explainability in algorithms engenders mistrust, particularly in academic evaluation and robotic administrative processes. Educators underscore the necessity for audits and control mechanisms that ensure the decisions made by AI are comprehensible and substantiated. This perception suggests the development of policies that would strengthen the accountability and traceability of algorithms used in higher education.

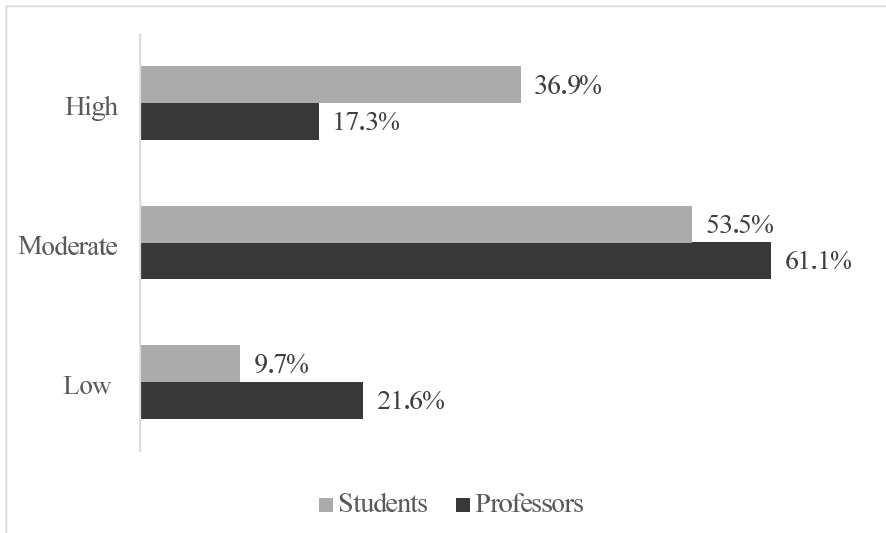


Fig. 2 Perception of AI transparency and explainability. Source: Data obtained from the questionnaire applied

Academic Responsibilities and Freedoms

This dimension analyses the effect of AI on autonomy and the decision-making process within educational institutions. It incorporates the examination of responsibility the event of errors or undesirable consequences derived from AI, as well as the assurance that these systems do not restrict the freedom of thought, research, and teaching in the university environment. Figure 3 shows the perceptions of the use of AI in terms of academic responsibilities and freedoms within Peruvian universities.

The data reveal a concern regarding academic autonomy in the implementation of AI. A considerable proportion of respondents, comprising 45.7% students and 51.9% professors, exhibited a moderate perception of the utilization of AI in teaching. Conversely, 43.7% of students and 17.9% of professors have expressed a high degree of concern that the integration of AI in pedagogical practices may potentially compromise academic freedom and the autonomy of faculty members in making educational decisions. Furthermore, concerns have been raised regarding the allocation of responsibility in instances of errors generated by these systems, underscoring the necessity for the establishment of control and supervision units in their utilization. The prevailing sentiment in the responses indicates a demand for enhanced clarity regarding the ethical and legal standards that govern the use of AI in higher education.

Equity and Non-discrimination

This dimension explores the potential of algorithms to either perpetuate or mitigate discriminatory biases within the context of higher education. It analyses equity in the allocation of resources, opportunities, and student assessment through AI, as well as the prevention of algorithmic biases that may affect vulnerable groups. It seeks to ensure that AI systems do

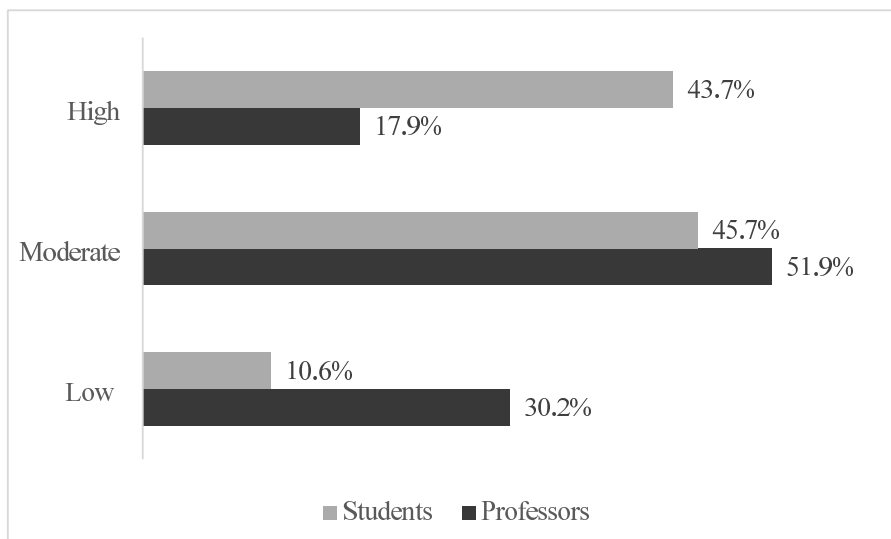


Fig. 3 Perceptions regarding academic responsibilities and freedoms in the use of AI. Source: Data obtained from the questionnaire applied

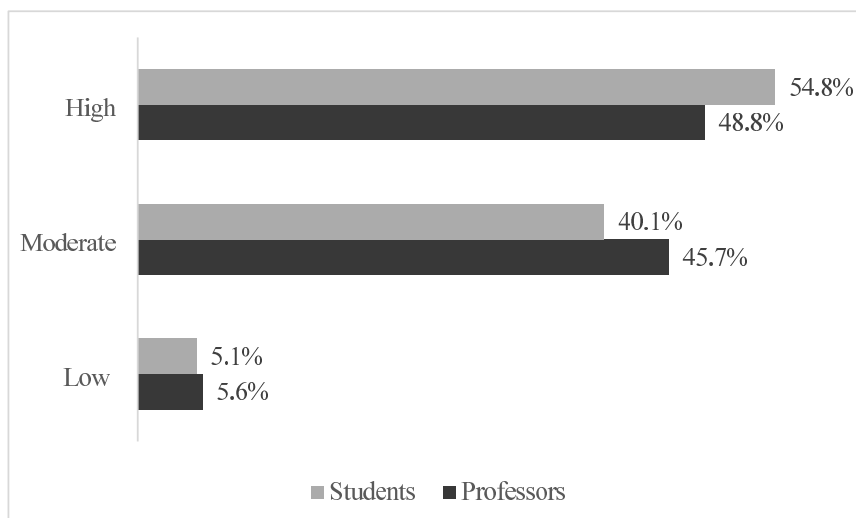


Fig. 4 Perceptions about equity and non-discrimination in the use of AI. Source: Data obtained from the questionnaire applied

not replicate pre-existing inequalities in access to and use of technology. Figure 4 describes how university stakeholders perceive equity and non-discrimination in the use of AI within universities.

The data presented in this figure indicates a modest degree of consensus between students (40.1%) and faculty (45.7%), with a notable disparity between faculty members (48.8%) and students (54.8%), who express a higher degree of confidence in AI systems' potential to

perpetuate discriminatory biases, particularly in academic evaluation and the distribution of educational resources. The results underscore the significance of algorithms being designed and implemented with equity criteria to prevent the exclusion of certain student groups. Furthermore, educators have articulated apprehensions regarding the absence of regulatory frameworks that would guarantee the impartiality of AI systems within the academic setting. These findings suggest that it is imperative to develop strategies to mitigate algorithmic biases in education.

Well-being and Social Benefit

It considers the impact of AI on the global well-being of the academic community and its contribution to collective well-being. It analyzes whether AI systems applied in universities increase the quality of education, reduce the administrative burden, or provide considerable advantages to students and professors. It also addresses the balance between automation and human interaction in the educational process. Figure 5 shows the perceptions about the well-being and social benefits that AI has within Peruvian universities.

The results of the study indicate that 49.7% of students and 56.8% of teachers exhibited a moderate perception of AI, perceiving it as a facilitator of improved welfare and social benefits. Conversely, 25.9% of professors and 43.3% of students exhibited a high perception of AI. While certain individuals acknowledge the advantages of such automation, including the potential enhancement of pedagogical practices and the streamlining of administrative operations, others voice concerns regarding the potential consequences, such as the dehumanization of education and an excessive reliance on technology. The graph indicates that a considerable set of participants consider that AI can contribute to educational development, as long as its application is ethical and respectful of academic values. This underscores the significance of maintaining a balance between automation and human interaction in the learning process.

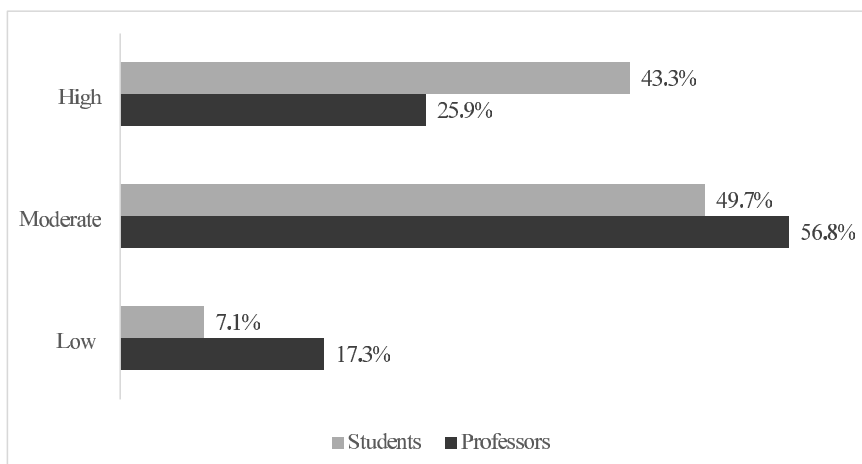


Fig. 5 Perceptions about the welfare and social benefits of using AI. Source: Data obtained from the questionnaire applied

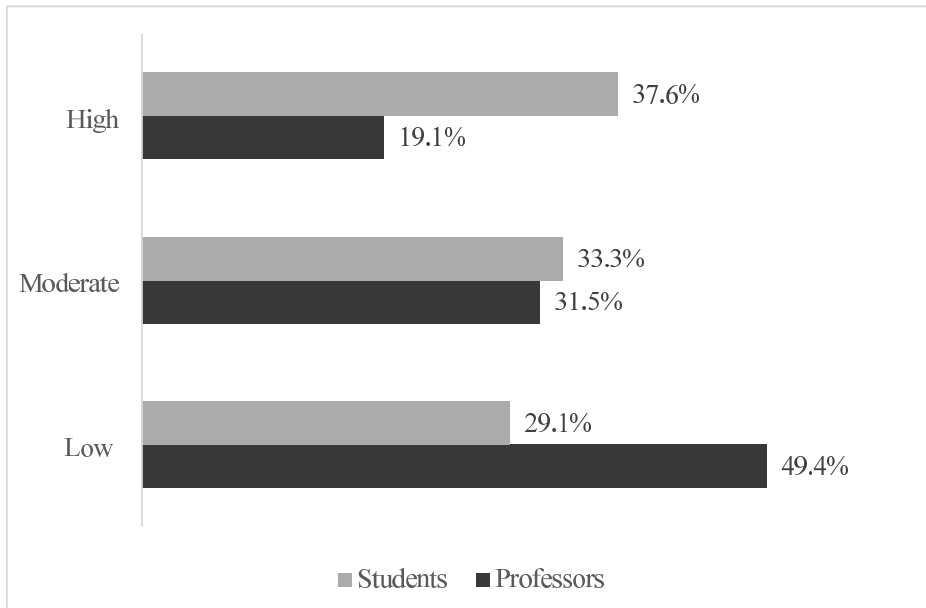


Fig. 6 Perceptions about AI training and literacy. Source: Data obtained from the questionnaire applied

AI Training and Literacy

This dimension explores the extent of knowledge and training that students and professors possess regarding the ethical utilization of AI. The incorporation of demands for educational programs on the ethics of AI, digital skills, and critical understanding of algorithmic systems is imperative. It emphasizes the training of the academic community to ensure informed and responsible interactions with AI within the context of education. Figure 6 describes how professors and students perceive training and literacy on AI in Peruvian universities.

The Fig. 6 indicates a substantial absence of training on AI in the university environment. According to the findings of the survey, 56.8% of the faculty members surveyed and 49.7% of the students reported that they have not received adequate training in the use and ethical consequences of artificial AI. Consequently, it is recommended that institutions of higher education implement digital literacy programs that facilitate the comprehension of these technologies among students and professors. Furthermore, the findings indicate that individuals with a robust understanding of AI tend to hold a favorable opinion of it, underscoring the significance of ongoing training in this domain.

Discussions

The findings reveal that a significant proportion of students and faculty members perceive an absence of clear policies regarding the privacy and data protection in the use of AI at their universities. This perception is consistent with the findings reported by Herrera (2022) and Mauti and Ayieko (2025), who caution that the absence of regulatory oversight can result

in the exploitation of personal data and the emergence of digital security risks. In a similar vein, Gupta and Nyamapfene (2025) contend that stringent ethical standards must govern the utilization of AI in educational settings to forestall excessive surveillance and the improper use of information. Likewise, Swist et al. (2024) emphasize the need to co-design ethical frameworks that safeguard student privacy, an issue that remains insufficiently addressed in the Peruvian context. This finding is indicative of a more extensive problem, as evidenced by studies conducted in Jordan Al-Ani et al. (2024) and India Annamalai et al. (2025) which demonstrate the prevalence of such challenges in other regions. This observation underscores the necessity for the establishment of regional and collaborative ethical frameworks.

Another key finding refers to the perception of opacity in the algorithms used in universities. This concern is consistent with the outcomes of studies conducted in analogous environments, including universities in Southeast Asia (Guan et al., 2025) and the Middle East (Al-Ani et al., 2024), where challenges related to auditing and comprehending algorithmic systems have been identified. As posited by Protopapa and Idris (2025), the absence of transparency constitutes an impediment to the efficacious integration of AI in educational settings. As Gallent et al. (2023) emphasize, the absence of explainability may engender hidden biases, which can have a deleterious effect on student assessment and performance. Furthermore, Bozkurt et al. (2021) and Kraemer et al. (2011) observe that algorithms are not impartial, as they incorporate implicit values that have the potential to amplify systemic biases. Černý (2024) asserts the pivotal role of digital literacy in addressing algorithmic transparency, facilitating the comprehension of professors and students regarding the boundaries and ramifications of these technologies.

Concerns regarding the impact of AI on educational autonomy have been expressed by both students and professors. Muwagunzi et al. (2024) have observed that the integration of automated platforms within educational settings has engendered a multitude of uncertainties, chiefly concerning the conventional teaching role and the management of curricular content. This ambiguity may result in the substitution of key pedagogical functions by algorithmic systems that are inadequately contextualized within cultural frameworks. This finding aligns with the observations documented by Chiu and Chai (2020), who noted instructor resistance to AI, perceiving it as a threat to their teaching autonomy. Dabis and Csáki (2024) and (Zawacki-Richter et al., 2019) concur on the necessity to safeguard academic autonomy and institute accountability mechanisms in the event of erroneous decisions made by automated systems.

The research also highlights concerns regarding potential algorithmic biases that could compromise educational equity. In nations such as India and Morocco, studies have documented instances of algorithmic practices that exhibit a replication of gender, class, and ethnic inequalities (Ferhataj et al., 2025; Kamali et al., 2024). These issues, related to the homogeneity of training data and the technocratic design of AI tools, contribute to the exclusion of vulnerable student groups. As asserted by Alonso-Rodríguez. (2024) and Caja et al. (2024) the absence of regulatory oversight can serve to exacerbate existing structural disparities. Conversely, Saleem et al. (2024) advocate for the ethical oversight of these technologies, promoting equity principles from the design stage. These challenges underscore the imperative for governance frameworks that guarantee the inclusive and ethical design of AI systems.

In light of these findings, it is imperative to acknowledge that ethical challenges in the implementation of AI in higher education should not be analyzed exclusively from a

national standpoint. The observed parallels between the perceptions collected in Peru and the documented evidence from other regions lend credence to the proposition of formulating regional governance frameworks. Initiatives such as the Pretoria Declaration in Africa (UNESCO, 2024) or India's Responsible AI for All strategy, developed by NITI Aayog (2021) offer conceptual references that, despite responding to different socio-political and technological contexts, can inspire Latin American efforts toward ethical regulation. Embracing South-South cooperation on AI ethics, particularly in the education sector, has the potential to facilitate the establishment of more context-sensitive and effective mechanisms for governance, training, and responsible innovation.

Nonetheless, the majority of participants acknowledged the advantages of automating administrative tasks and personalizing learning. However, a minority expressed concerns regarding the potential dehumanization of education. While respondents agreed that AI optimizes teaching, there is concern about the potential dehumanization of higher education. This dilemma has been widely explored by Lee and Shim (2024) and Amén et al. (2024), who highlight that while AI provides sophisticated learning instruments, it is incapable of supplanting empathy and human interaction. Ortega et al. (2024) and Annamalai et al. (2025) have noted that excessive reliance on AI has the potential to impede the cultivation of students' critical thinking and creative abilities. Conversely, Kayali et al. (2023) and Miao et al. (2021) argue that AI is a valuable tool when used complementarily. However, Muwagunzi et al. (2024) emphasize that the efficacy of this approach is contingent upon the implementation of regulatory frameworks aimed at balancing automation and human interaction.

A notable finding in Peruvian universities is the recognition of a training gap in the use of AI. This study reveals that a significant number of students and professors believe they have not received adequate training, resulting in a gap in digital literacy. This issue has been previously discussed by Kit et al. (2025), who argue that AI education is essential for its ethical and efficient use. As Černý (2024) also emphasizes that a lack of knowledge can lead to misuse or uncritical dependence. Guan et al. (2025) and Summers et al. (2024) posit that universities should integrate AI education into their curricula and provide training to both students and professors. Additionally, Barrios (2023) cautions that in the absence of adequate digital literacy, students may depend on AI for text production, neglecting to cultivate their own analytical and argumentative capabilities.

In this context, the findings underscore the necessity for explicit institutional policies and robust training strategies to regulate AI in higher education, as discussed by Alonso-Rodríguez (2024) and Dabis and Csáki (2024). The integration of AI into educational settings is a rapidly evolving field, necessitating the establishment of guidelines to ensure its ethical, transparent, and fair implementation. Tsamados et al. (2022) and Mökander & Floridi (2021) have noted that audits of AI systems are imperative for ensuring reliability and preventing bias. Furthermore, the necessity to enhance AI-related training, as highlighted by Černý (2024) and Kit et al. (2025), is paramount to ensure the adequate preparation of future professionals. In light of these findings, it is imperative to emphasize that ethical challenges in the implementation of AI in higher education must be addressed through a comprehensive and collaborative international perspective.

Conclusions

The results emphasize the need for universities to take specific actions to address the ethical challenges of AI in higher education. Implementing regulatory frameworks, audit procedures, and digital literacy training programs is essential to optimizing the benefits and mitigating the risks identified in this study.

The study emphasizes the importance of addressing the ethical challenges associated with using AI in higher education, particularly at Peruvian universities. AI has tremendous potential to improve teaching and learning, as well as administrative processes. However, applying AI without a solid regulatory framework will consequently trigger concerns about data privacy, algorithmic bias, a lack of clarity in decision-making processes, and inequity in access to education.

From ethical and academic perspectives, incorporating AI into teaching requires honoring principles such as academic autonomy, equity, and responsibility. This ensures that automation does not replace critical thinking or human participation in the learning process.

In order to ensure the ethical and effective use of AI in higher education, clear institutional regulations are required. These regulations should address algorithmic transparency, data privacy, and equitable access to technology. Additionally, digital literacy programs and AI ethics training should be developed for students and professors to encourage the informed and critical use of these tools.

Examining the impact of AI on teaching is imperative, as is correcting possible biases or disadvantages in its application through monitoring mechanisms and audits. From a social perspective, we propose promoting technological inclusion to prevent disparities in access to these tools, reinforcing the protection of personal data through robust security protocols, and establishing alliances with the public and private sectors to integrate AI training in the workplace. This approach is expected to better prepare students for the challenges of digital transformation.

The findings of this study are situated within the Peruvian context but reflect ethical concerns identified in higher education systems worldwide. Issues such as data privacy, algorithmic bias, a lack of transparency, and unequal access to AI are not unique to Peru but are part of a broader set of challenges shared by various regions worldwide. This convergence suggests that the ethical challenges experienced by Peruvian students and faculty are part of a global landscape in which AI governance still has regulatory gaps and institutional tensions. Positioning the findings within the international discourse enhances the study's relevance and underscores the importance of developing collaborative, context-sensitive ethical frameworks to guide the responsible use of AI in higher education.

This study provides valuable insights into the ethical challenges of AI in higher education, but it has limitations. For example, the use of self-reported data may introduce social desirability bias or subjective interpretations, which could affect the accuracy of the findings. Additionally, reliance on convenience sampling based on voluntary participation may have introduced selection bias, limiting the representativeness of the sample and reducing the generalizability of the results to the wider university population. Future research should adopt more robust sampling strategies and complement survey data with qualitative methods or observational evidence to capture a more comprehensive and reliable picture of the ethical implications of AI in academic settings. Despite these limitations, the present study

meaningfully contributes to developing institutional policies that promote the ethical and equitable integration of AI in higher education.

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Data Availability The data is available upon request from the main author.

Declarations

Conflict of Interest The authors declare no conflict of interest.

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Authors and Affiliations

Yuri Reina Marín¹  · Omer Cruz Caro¹  · Angelica María Carrasco Rituay²  · Katia Araceli Guimac Llanos¹  · Doris Tarrillo Perez¹  · Einstein Sánchez Bardales³  · Judith Nathaly Alva Tuesta³  · River Chávez Santos⁴ 

✉ Omer Cruz Caro
omer.cruz@untrm.edu.pe

Yuri Reina Marín
yuri.reina@untrm.edu.pe

Angelica María Carrasco Rituay
angelica.carrasco@untrm.edu.pe; e.angiecarrasco@go.ugr.es

Katia Araceli Guimac Llanos
katia.guimac@untrm.edu.pe

Doris Tarrillo Perez
7346486221@untrm.edu.pe

Einstein Sánchez Bardales
einstein.sanchez@untrm.edu.pe

Judith Nathaly Alva Tuesta
judith.alva@untrm.edu.pe

River Chávez Santos
river.chavez@untrm.edu.pe

¹ Oficina de Gestión de la Calidad, Universidad Nacional Toribio Rodríguez de Mendoza de Amazonas, Chachapoyas, Perú

² Universidad de Granada, Granada, España

-
- ³ Escuela de Posgrado, Programa Doctorado en Ciencias para el Desarrollo Sustentable, Universidad Nacional Toribio Rodríguez de Mendoza de Amazonas, Chachapoyas, Peru
- ⁴ Facultad de Educación y Ciencias de la Comunicación, Universidad Nacional Toribio Rodríguez de Mendoza de Amazonas, Chachapoyas, Peru